Waiter

A self-hosted memory retrieval API - LongMemEval-S results for discussion

HYBRID RRF - RECALL@1
57.0%
+/-3.3 pp; 365 held-out questions

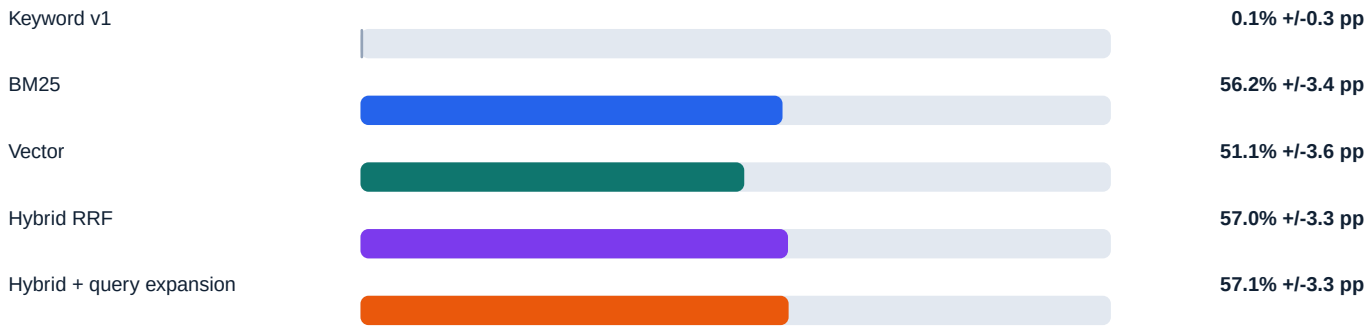
HYBRID RRF - MRR@20
93.0%
rank of first relevant session

HYBRID RRF - RECALL@20
98.8%
saturated at this candidate count

WARM RETRIEVAL P50
0.05 ms
in-process; excludes embedding

Recall@1 - Waiter retrieval configurations

Higher is better. The best observed result is query expansion at 57.1%; its +0.1-point difference over plain Hybrid RRF is within uncertainty and does not establish a real gain.



THE MEETING TAKEAWAY

Waiter moved from literal substring matching (Recall@1 0.1%) to BM25 retrieval (Recall@1 56.2%). Hybrid RRF reaches 57.0%. This demonstrates a useful, reproducible retrieval core—not that Waiter beats Supermemory end-to-end.

Ranking metrics (higher is better)

Configuration	Recall@1	Recall@10	Recall@20	NDCG@10	MRR@20
BM25	56.2%	95.2%	97.8%	90.4%	91.6%
Vector	51.1%	95.3%	98.3%	87.0%	86.6%
Hybrid RRF	57.0%	96.9%	98.8%	92.2%	93.0%
Hybrid + query expansion	57.1%	97.1%	99.2%	92.4%	93.1%

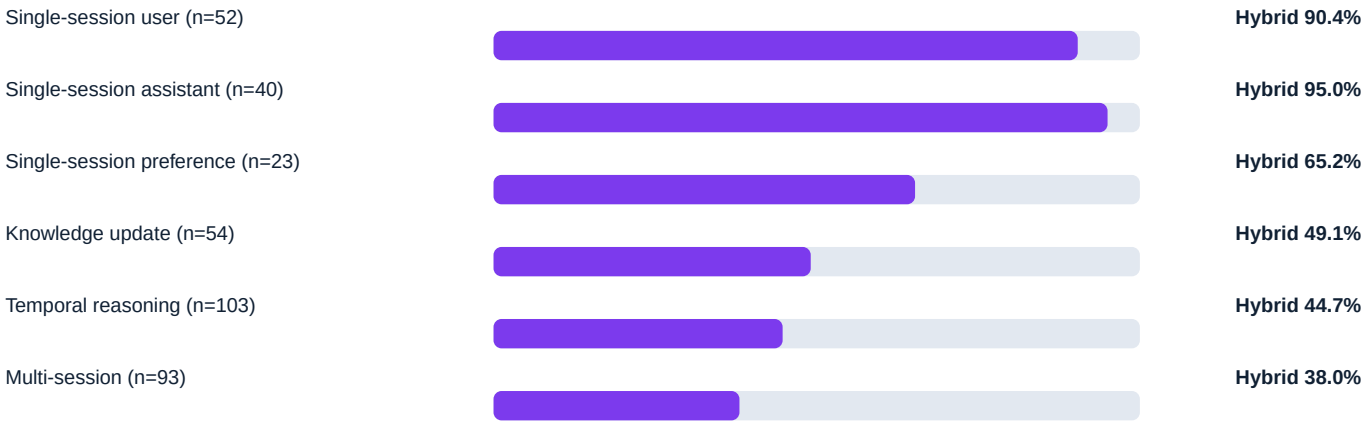
Dataset: 500 cleaned LongMemEval-S questions; 135 dev / 365 test, split by question-id hash. One indexed memory per session. Embeddings: local all-MiniLM-L6-v2. No answer model or LLM judge was run.

Where Waiter works - and where to be careful

Category results, public competitor claims, and what a fair comparison would require

Recall@1 by LongMemEval question category

BM25 and vector retrieval complement each other on some categories. Preference has only 23 test examples; read that result as an indication, not a firm ranking.



Category (n)	BM25 R@1	Vector R@1	Hybrid R@1
Single-session user (n=52)	92.3%	73.1%	90.4%
Single-session assistant (n=40)	100.0%	92.5%	95.0%
Single-session preference (n=23)	43.5%	69.6%	65.2%
Knowledge update (n=54)	50.0%	47.2%	49.1%
Temporal reasoning (n=103)	45.1%	38.3%	44.7%
Multi-session (n=93)	36.3%	33.0%	38.0%

WHAT WE CAN SAY ABOUT SUPERMEMORY

Supermemory's research report publishes 97% LongMemEval-S Recall@20 with aggregation; its GitHub README separately advertises 95% Recall@15 and 99.4% context reduction. These are vendor-published claims. Waiter's 98.8% Recall@20 is from a cleaned, session-retrieval run without the same stated aggregation protocol: it is not a head-to-head win or proof of parity. BM25 alone already scores 97.8% here, so Recall@20 is saturated.

Scope comparison

WAITER today: BM25 + optional vectors + RRF; caller writes text; in-memory Node/Express; retrieval only.
Supermemory public description: memory extraction, profiles, hybrid search/RAG, atomic memories, relations, temporal grounding.
Evaluation: Waiter retrieval results here; Supermemory vendor-published LongMemEval claims, not reproduced here.

FAIR NEXT STEP

A head-to-head needs the same dataset version, ingestion and aggregation protocol, answer model, judge prompts and token budget, plus per-question scores for a paired confidence interval. Waiter has not run LoCoMo, ConvoMem, or the LongMemEval QA half.

Sources: supermemory.ai/research/longmembench; github.com/supermemoryai/supermemory; github.com/xiaowu0162/LongMemEval. Raw Waiter run: bench/results/results.json.